RESEARCH STRATEGY

A. Significance

Genome-wide association studies (GWAS) are the principal study design used to investigate the genetic architecture of complex traits in humans. GWAS—which regress phenotype onto genotype with some controls such as genetic principal components—typically aim to identify direct genetic effects (DGEs, i.e., causal effects of alleles in an individual on that individual). Most downstream analyses using GWAS summary statistics, including estimation of heritability and genetic correlation[1,2] and inference of disease causes using Mendelian Randomization[3], also assume (usually implicitly) that the GWAS estimates are DGEs. Yet research from our group and others[1,2,4–9] has established that, across a broad range of phenotypes, GWAS estimates are affected by substantial confounding from indirect genetic effects from parents and other relatives (IGEs, i.e., effects of relatives' alleles that affect the individual through the environment), assortative mating (AM), and other complex intergenerational dynamics such as uncorrected-for population stratification (Figure 1a). The same confounding factors likely contribute to the drop in predictive power of polygenic predictors when applied across ancestries[10,11].

For many human traits and diseases, there is substantial evidence that genetic architecture is affected by complex intergenerational dynamics. For example, AM is widespread within and across many phenotypes, and cross-trait AM (xAM: assortment resulting in cross-trait correlations across mates) can lead to significant estimated genetic correlations even in the absence of pleiotropy[2] (Figure 1b). Based on preliminary results from a meta-analysis of family-based GWAS led by PI Young, we have shown that the genome-wide correlations between DGEs — as estimated by family-based GWAS[4,8] — and standard GWAS estimates (which we call "population effects") are far below one for many biomedical, psychiatric, social, and behavioral traits (Figure 1c).

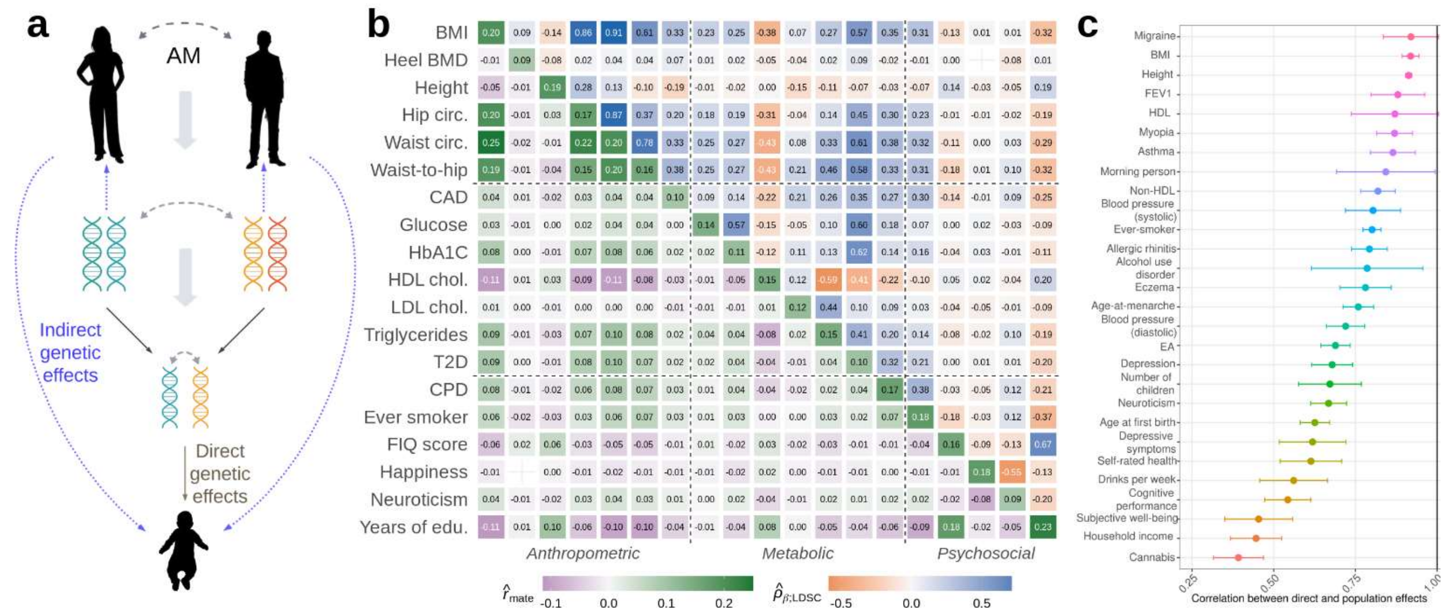

***Figure 1**. The causal pathways from parental genotype to offspring genotype includes not only direct genetic effects (DGEs), but also indirect genetic effects (IGEs) from parents (environmentally mediated effects of the parental genome), assortative mating (AM), and population structure (not depicted) (b) Cross-mate correlations (lower diagonal) are widespread and explain 74% of the variation in LDSC genetic correlation estimates (upper diagonal)[2] (c) Genome-wide correlations between DGEs (from family-based GWAS) and population effects (from standard GWAS); error bars give 95% C.I.s.*

For intuition on how such phenomena can influence genetic architecture and distort apparent relationships between genotypes and phenotypes, consider two traits that have been studied in large GWAS samples: educational attainment (EA) and height. If parental education affects offspring education through the rearing environment (an example of vertical transmission[12], VT), then alleles that affect parental EA will affect offspring EA (an example of IGEs), as will non-genetic components of parental EA. Moreover, EA and height are subject to xAM (both height and EA in one mate predict both traits in the other[2]; see Figure 1b). Thus, over successive generations, trait-increasing alleles for both traits become positively correlated with each other (within and across traits), leading to long-range linkage disequilibrium and biased estimates of the genetic architecture of both traits. Because xAM and VT are widespread[1,8,13,14], family-based methods are needed to make robust inferences about biology, and an understanding of the intergenerational dynamics (AM, IGEs, etc.) is necessary to model intergenerational inequalities accurately. This project aims to develop simulation, modeling, and estimation tools that will enable the human genetic community to delineate the contribution of DGEs, IGEs, and AM to human traits as well as to test the sensitivity of existing methods to these and other phenomena.

A.1. (Aim 1) Development of theory and robust methods for estimating indirect genetic effects, with applications to multi-ancestry samples and phenotypes from multiple datasets.

Understanding the mechanisms through which the family environment affects offspring outcomes is key to understanding intergenerational inequalities in health and education. One mechanism is parents through their role in shaping the rearing environment. However, separating environmentally mediated effects of parents from other factors influencing is incredibly challenging. One approach to this problem is to establish evidence for IGEs from parents, which are consequences of environmentally mediated effects of parents (as in VT models). Most existing estimates of IGEs are based on a method by Kong et al.[13] (co-authored with PI Young). Kong et al. found that the PGI for EA constructed from *non*-transmitted alleles predicted offspring EA, giving evidence for parental IGEs. Subsequent studies[15,16] have similarly interpreted associations between offspring phenotypes and non-transmitted parental alleles as parental IGEs. However, AM, even without IGEs, can lead to associations between offspring phenotypes and non-transmitted alleles by inducing correlations between transmitted and non-transmitted alleles[17], and studies have suggested that AM and residual population stratification have inflated estimates of parental IGEs[18,19]. All existing methods of estimating IGEs/VT suffer from critical shortcomings:

• **Lack of a general theoretical framework that includes the joint effects of DGEs, IGEs, and AM**. Previous theoretical approaches based on VT models make strong assumptions about the mechanism of AM and do not explicitly model IGEs of variants so are difficult to connect to genomic data analyses.

• **Invalid assumptions**: previous methods to account for AM make strong and likely invalid assumptions about the parental phenotype through which VT acts[12,20], the correlation between DGEs and IGEs[21], and heritability[12,21].

• **IGE studies have been of European genetic ancestry samples.** As IGEs and AM lead to confounding in GWAS and polygenic prediction analyses, evaluating how they vary across populations is critical for understanding the factors that influence portability of polygenic prediction across ancestries[10,11].

A.2. (Aim 2) Development of advanced simulation tools and open data library infrastructure.

We propose to develop efficient, flexible simulation tools together with an open simulation library and benchmarking infrastructure for examining broadly defined phenotypic & genetic architectures. Using these tools, we will assess existing methods for association studies, variance components analysis, and polygenic prediction, while creating robust infrastructure for the development of novel methods. We will address several critical deficiencies in existing simulation approaches:

• **Complex trait architecture has been narrowly defined** in terms of DGE distributions as a function of locus-level annotations (i.e., allele frequency, local LD, and/or biological function)[22–24], disregarding the considerable impacts of intergenerational dynamics, which alter the joint distribution of variants, effects, and environmental factors.

• **Only overly simplistic forms of intergenerational dynamics have been examined**. Previous work has assumed that no more than two traits are involved in mate choice, with most assuming univariate AM[2,25,26]. Empirical data shows that xAM is multi-dimensional and involves many disease-relevant phenotypes (Figure 1b).

• **Perturbations of assumptions are generally examined in isolation**. Multiple assumptions typically made by human genetics methods will be simultaneously invalid for many traits of interest: e.g., EA is correlated with many health-related phenotypes across mates, and is thought to be affected by IGEs, yet no studies have examined the joint impact of these factors on statistics such as genetic correlation. As a result, uncertainty about basic questions such as whether there is substantial pleiotropy between EA and health phenotypes remains[8].

• **Existing approaches to sensitivity analysis typically take the form of bespoke simulation scripts that are fragmented across research groups**. This often results in inefficiencies such as independent research recreating similar simulations[23,24,27]. Beyond inefficiencies due to duplicated efforts, many methods researchers will not examine relevant scenarios at all due to the burden of bespoke implementation, and the choice of scenarios to study may inadvertently tilt the conclusions in favor of whatever method is being proposed.

B. Innovation

This application directly challenges existing research paradigms by developing and applying novel theory and methods for modeling and estimation of IGEs and xAM, as well as their consequences for current methods. Carefully delineating the complex forces that contribute to genetic architectures and phenotypes will represent a major conceptual advance. Our project will also innovate in producing efficient algorithms and easy-to-use software for simulating complex intergenerational dynamics and estimating DGEs, IGEs, and AM.

In **Aim 1,** we enable robust estimation of DGEs and IGEs and provide, for the first time, estimates of IGEs and DGEs for a range of phenotypes across multi-ancestry populations by:

• Developing a novel theoretical framework for joint impact of DGEs, IGEs, and AM. We generalize classical theory of how AM affects the DGE component from Crow and Felsenstein[28] to include IGEs. Unlike previous work[29], we do not rely on a particular model of AM, do not make assumptions about parental phenotypes through which IGEs operate, and we model arbitrary correlations between DGEs and IGEs. By modeling DGE and IGE components directly, our model is more easily connected to analysis of genomic family data than classical theory, enabling more robust inference of parameters than previous approaches under weaker assumptions.

• Developing methods for estimating IGEs and heritability accounting for AM in two-generation models. Most approaches to estimating IGEs are subject to significant bias from population stratification, and confounding with DGEs through AM[14]. Attempts to address this shortcoming[12,21], rely on unrealistic assumptions, do not account for AM bias in heritability estimates[1,30], or require extensive additional information. Our proposed method relaxes many assumptions, giving valid answers under a wider range of scenarios. Furthermore, our proposed method enables estimation of the correlation between parents' underlying DGE components, enabling adjustment for AM-induced bias in twin and family estimates of heritability[30].

• Enabling robust estimation of IGEs using three-generation models. While our two-generation approach represents a substantial advance, it does not account for potential bias due to population stratification. We create a three-generation approach which uses meiosis in grandparents to fully remove confounding due to population stratification, and most of the confounding due to AM. To overcome the lack of fully genotyped three-generation pedigrees, we impute missing grandparental PGIs using an extension of methodology previously developed by PI Young[4]. This is an important innovation that enables three-generation analyses despite a lack of fully genotyped three-generation pedigrees.

• Application of robust methods for estimating IGEs to multi-ancestry cohorts. Previous empirical work on IGEs has focused on European genetic ancestry samples. An important innovation of our project is analyzing multi-ancestry samples from the Mexico City Prospective study (MCPS)[31] and Taiwan Biobank[32], which are <u>public datasets</u> with thousands of genotyped sibling and parent-offspring pairs as well as a wide range of phenotypes. We will apply our proposed methods to MCPS and Taiwan Biobank, <u>providing the first comprehensive estimates of IGEs in non-European genetic ancestry samples</u>.

<u>In **Aim 2** we will identify biases in existing approaches to estimating genetic architecture and improve future development of robust methods by:</u>

• Creating software for multivariate forward time simulation of complex intergenerational dynamics. Existing dynamic simulation tools are primarily aimed at long timescale population genetic simulations[33,34]. Existing forward time simulators capable of representing xAM, IGEs, and other short timescale dynamics (e.g. AlphaSimR[35], GeneEvolv[36]) are limited in scope and do not scale to biobank level analyses. Here, we develop the eXtensible Forward Time SIMulator (*xftsim*) library to enable 1) realistic biobank-scale haplotype data; 2) complex genetic and phenotypic architectures with cross-generational dependences; 3) high-dimensional assortative mating; 4) spatial and ancestry-based gradients in environment and mating; and 5) incorporation of -omics and other auxiliary phenotype data.

• Deriving and implementing efficient one-shot genotype & phenotype simulation and theory. Existing one-shot simulators (i.e., without multigenerational components) cannot represent the long-range LD generated by xAM[37–39]. We leverage the linear time multivariate Bernoulli random sampling algorithm (*rb_dplr*) developed by Co-I Border[40] by extending classical quantitative genetic theory on AM — which focused on at most two phenotypes at equilibrium[17,41–43] — to equilibrium and disequilibrium multivariate xAM. This will allow for efficient generation of genotype & phenotype data for important scenarios targeted in Aim 1.1.

• Creating a simulation and benchmarking platform for characterizing the robustness of current and future methods to complex intergenerational dynamics. The lack of shared simulation and benchmarking infrastructure means our understanding of human genetics methods across the spectrum of relevant architectures is spread across a fragmented literature, making it challenging to compare performance across methods and scenarios and to identify weaknesses common across currently available methods. Here we develop an open platform enabling realistic, reproducible assessment of existing and future human genetics methods.

C. Approach

C.1. The Research Team. A key strength of this proposal is the varied nature of the research team, it is composed of men and women at early and mid-career stages, from multiple geographic locations, and of different disciplinary backgrounds. Collectively we have published 51 co-authored papers demonstrating a track record of success on this topic attributed in part to the interdisciplinary backgrounds required for this work:

PI Alexander Young (Assistant Professor) is a statistical geneticist with a record of high-impact publications on family-based methodologies[4,14,21,44], gene-gene and gene-environment interactions[45–47], and heritability estimation[44,48]. He leads an international consortium for family-based GWAS analyses.

Co-I Daniel Benjamin (Professor) is an economist with expertise in GWAS of behavioral and social phenotypes[14,49–57], incorporating genetics into social science[58–60], and ethics across genetics and social sciences[61–63]. He is a co-founder of the Social Science Genetic Association Consortium, which coordinates multidisciplinary collaborations across cohorts with large-scale genomic data.

Co-I Noah Zaitlen (Professor) has a background in computer science and has developed widely used software tools for multi-ethnic genetic methods[64–68] and genetic architectures[2,69–72].

Co-I Richard Border (Assistant Professor) has a background in psychology and applied math, with a record of publications on (x)AM[2,40,73], psychiatric genetics[1,74–78], and heritability & genetic correlation estimation[1,2,79].

Co-I Nanibaa' Garrison (Associate Professor) is a bioethicist with expertise on the societal implications of genetic studies[80,81], especially as related to the inclusion of non-European-genetic-ancestry communities in genetics research[82–85].

Co-I Aaron Panofsky (Professor) is a sociologist who studies the social and political dimensions of genetics research[86,87], as well as its translation into public understanding[88,89] through media.

We have enlisted two consultants, **Loic Yengo** and **Mashaal Sohail**. Loic Yengo is a statistical geneticist with expertise in assortative mating[73,90–92] and analysis of multi-ancestry data[10,93]. Mashaal Sohail — who will assist in our analyses of the Mexico City Prospective Study — is an expert in the population genetics of Mexico[94,95] and the use of ancestry in human genetics research[95–98].

C.2. Sex Specific Analysis. All analyses will be additionally conducted separately in males and females, when possible, to test for sex-specific effects. Simulation tools are designed to allow for sex-specific trait architectures.

***C.3. Aim 1:* Development of theory and robust methods for estimating indirect genetic effects with multi-ancestry application.**

***C.3.1. Overview*.** We first derive a theoretical framework for the joint impact of DGEs, IGEs, and AM at equilibrium. This theoretical framework allows us to understand the results of two-generation PGI analyses and to adjust IGE and heritability estimates for bias due to AM. The simplest two-generation PGI analysis performs a joint regression of samples' phenotypes onto their own PGI and their parents' combined PGI: $Y_{ij} = \delta_{\text{PGI}}\text{PGI}_{ij} + \alpha_{\text{PGI}}\text{PGI}_{\text{par}(i)} + \epsilon_{ij}$, where $\text{PGI}_{ij} = \sum_{l=1}^{L} w_l g_{ijl}$; $\text{PGI}_{\text{par}(i)} = \sum_{l=1}^{L} w_l (g_{p(i)l} + g_{m(i)l})$; w_l is the weight of variant l; and g_{ijl}, $g_{p(i)}, g_{m(i)}$ are, respectively, the mean-normalized genotypes of offspring j, the father, and the mother in family i. Because the offspring and parental PGIs use the same weights, the expectation of the offspring PGI given the genotypes of the parents is $\text{PGI}_{\text{par}(i)}/2$, with variation around this expectation due to random Mendelian segregations. This implies that δ_{PGI}, which we call the 'direct effect' of the PGI, reflects only DGEs[4,14].

However, as Kong et al.[21] showed, when the PGI does not capture all of the heritability and there is AM, α_{PGI}, which we call the average non-transmitted coefficient (NTC) of the PGI, includes a contribution from correlation between the PGI and the heritable component of the phenotype that would be uncorrelated with the PGI in a random-mating population (but is correlated with the PGI due to AM)[21].To give some intuition for this, consider a PGI constructed from variants on chromosome 1. In a random-mating population, the PGI is uncorrelated with genetic variants on other chromosomes. However, if there is AM, this will induce correlations between the chromosome 1 PGI and the rest of the genetic component of the phenotype from the other chromosomes. This correlation will contribute to association between phenotype and PGI but will not contribute to δ_{PGI} because chromosomes segregate independently within-family; instead, the correlation contributes to α_{PGI}. This logic also applies to situations where the PGI does not capture all of heritability due to incomplete genotyping (e.g. missing or poorly imputed rare variants) and/or due to imperfect correlation of the weights with the true DGEs[12,21].

We develop theory that enables interpretation of two-generation PGI analyses and derive a method that enables adjustment of IGE and heritability estimates for bias due to AM. To overcome some limitations of the two-generation approach, we develop three-generation genetic models for estimation of IGEs. We do this using an extension of the imputation methodology[4] previously developed by PI Young to overcome the lack of fully genotyped three-generation pedigrees. We then detail the datasets and populations in which we will apply our methods.

C.3.2. Theoretical framework combining indirect genetic effects and assortative mating. Crow and Felsenstein[28] gave derivations of theoretical results on AM using elementary methods. We generalize Crow and Felsenstein's approach (which considered only DGEs) by including parental IGEs in the following model: $Y_{ij} =$

$\Delta_{ij} + \eta_{p(i)} + \eta_{m(i)} + \epsilon_{ij}$; where ϵ_{ij} is the residual with variance σ_ϵ^2; and the components are given by $\Delta_{ij} = \sum_{l=1}^{L} \delta_l g_{ijl}$; $\eta_{p(i)} = \sum_{l=1}^{L} \eta_l g_{p(i)l}$; $\eta_{m(i)} = \sum_{l=1}^{L} \eta_l g_{p(i)l}$; where δ_l and η_l are, respectively, the DGE and parental IGE of variant l. We assume that paternal and maternal IGEs are equal, but we have shown that results given here apply when they are unequal with η_l interpreted as the average parental IGE. By generalizing the approach of Crow and Felsenstein, we express the equilibrium phenotypic variance as a function of the random-mating variance components (as defined by PI Young in relation to the RDR method of estimating heritability[44]) and the equilibrium correlations between the parents' DGE and IGE components (Figure 2): $\mathrm{Var}(Y_{ij}) = \frac{v_g}{1-r_\delta} + \frac{1+r_\eta}{1-r_\eta} v_{e\sim g} + \left(r_{\delta\eta}^{c} + r_{\delta\eta}^{\tau}\right)\sqrt{\frac{2v_{e\sim g}v_g}{(1-r_\eta)(1-r_\delta)}} + \sigma_\epsilon^2$, where v_g,$v_{e\sim g}$ are, respectively, the variances due to DGEs and IGEs under random-mating. The first two terms on the right-hand side (RHS) give the variances due to DGEs and IGEs at equilibrium, with the variance due to DGEs at equilibrium inflated by a factor of $(1 - r_\delta)^{-1}$ relative to random-mating, in agreement with classic results[17,28]. The third term on the RHS gives the variance due to covariance between DGE and IGE components at equilibrium, which can be non-zero even when DGEs and IGEs are uncorrelated due to the correlations induced by AM (Figure 2). An important implication of this is that, even if DGEs and IGEs are uncorrelated, a PGI constructed from DGE estimates is likely to be correlated with the IGE component under AM.

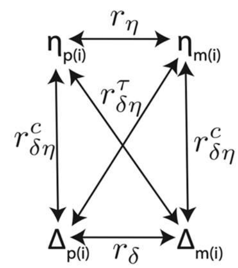

***Figure 2.** Equilibrium correlations between parents' DGE and IGE components.*

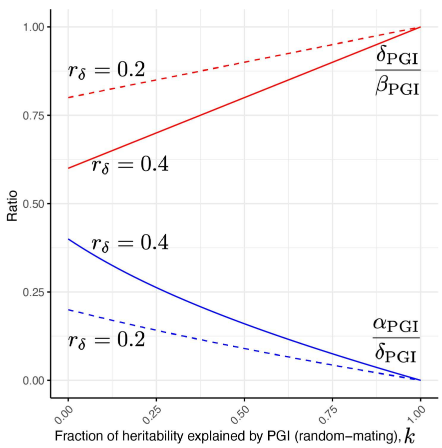

***Figure 3.** Expected results from PGI analysis under assortative mating in the absence of IGEs.*

We derive novel results giving the expected coefficients in two-generation PGI analysis under AM at equilibrium in the absence of IGEs (Figure 3). We assume that the PGI is constructed from unbiased DGE estimates and that there is no population stratification. Since Kong et al., attention has been given to the ratio between the average NTC and the direct effect of the PGI, $\alpha_{\mathrm{PGI}}/\delta_{\mathrm{PGI}}$, as a measure of the impact of IGEs relative to DGEs. We show this ratio depends upon the fraction of heritability the PGI would explain under random mating, k; and the equilibrium correlation between parents' DGE components, r_δ. Figure 3 shows $\alpha_{\mathrm{PGI}}/\delta_{\mathrm{PGI}}$ can be substantial when AM is strong and the PGI captures only a minority of the heritability (true for most current PGIs). We give a novel result for the ratio between direct and population effects of a PGI, where the population effect is defined by the univariate regression of phenotype onto PGI: $\beta_{\mathrm{PGI}} = \mathrm{Cov}(Y_{ij}, \mathrm{PGI}_{ij})/\mathrm{Var}(\mathrm{PGI}_{ij})$. The ratio $\delta_{\mathrm{PGI}}/\beta_{\mathrm{PGI}}$ expresses how much smaller the PGI-phenotype association is within-family, which is 1 minus how much the association 'shrinks' within-family[14,99]. Figure 3 shows that this shrinkage can be substantial due to AM (without IGEs). For example, for EA, AM is known to be strong[100]. Since current EA PGIs capture only a small fraction of the heritability of EA — with the latest EA PGI estimated to explain only 7% of the heritability estimated by twin studies[14] — this implies that values in the range of those observed in analyses of EA PGIs[14,15] ($\alpha_{\mathrm{PGI}}/\delta_{\mathrm{PGI}}$=0.3-0.5) could be due to AM alone.

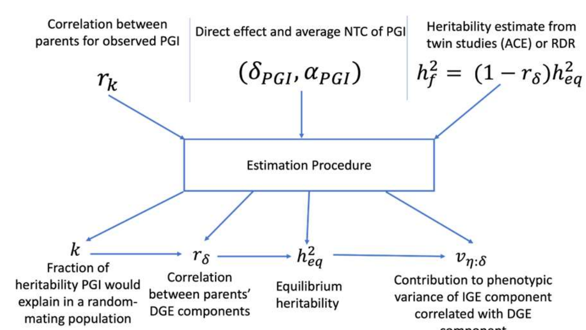

***Figure 4.** Procedure for estimating IGEs and heritability account for assortative mating.*

C.3.3. Methodology for estimating indirect genetic effects and heritability accounting for assortative mating. We outline a methodology that enables estimation of IGEs accounting for AM. We first estimate the fraction of heritability the PGI would explain in a random-mating population. This is complicated by the fact that AM biases most methods of estimating heritability, including the ACE model from twin studies[1,30]. We show that methods for estimating heritability based on variation in relatedness due to Mendelian segregation — sib-regression[101] and RDR[44] — estimate $h_f^2 = (1 - r_\delta)h_{\mathrm{eq}}^2$, i.e. the heritability at equilibrium, h_{eq}^2, shrunk by a factor of $(1 - r_\delta)$, the same as the ACE model[30]. We derive a series of non-linear estimating equations that take as inputs estimates of δ_{PGI}, α_{PGI}, h_f^2, and $r_k = \mathrm{Corr}\left(\mathrm{PGI}_{p(i)}, \mathrm{PGI}_{m(i)}\right)$, the correlation between parents' PGIs: the procedure (Figure 4) outputs k, r_δ, h_{eq}^2, and $v_{\eta:\delta}$, the contribution to phenotypic variance from the IGE component that is correlated with the DGE component — a measure of the overall contribution of IGEs to phenotypic variation. We give standard errors and confidence intervals using the Delta Method. We have implemented a prototype of this procedure in the software package *snipar*, which already has functions for performing two-generation PGI analyses[4]. This procedure is useful beyond estimation of IGEs:

it estimates the correlation between the unobserved full DGE components of the parents, giving insight into AM and enabling heritability estimates to be adjusted for bias due to AM, useful even for traits unaffected by IGEs.

C.3.4. Preliminary two-generation results. We first tested our procedure in simulations (Figure 5) with varying strengths of AM and varying correlations between DGEs and IGEs. With a PGI constructed from DGE estimates with $k \approx 0.5$, the procedure gives accurate estimates of h^2_{eq} and $v_{\eta:\delta}$. However, when the DGEs are estimated with greater noise, we found that the estimation procedure can become unstable.

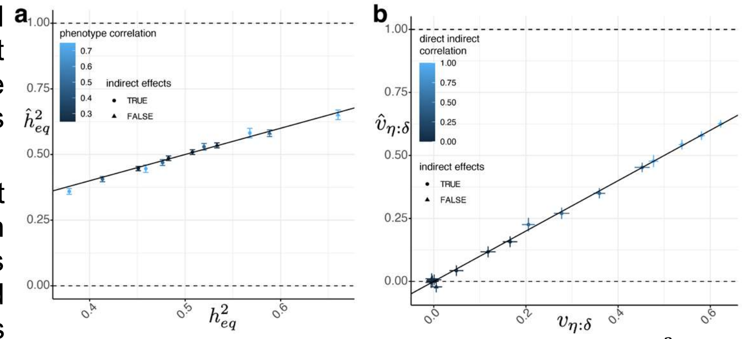

Figure 5. True versus estimated a) equilibrium heritability, h^2_{eq}, and b) variance due to the IGE component correlated with the DGE component, $v_{\eta:\delta}$. Results from applying our procedure to a DGE PGI that explains 50% of the heritability in a random-mating population.

Due to a lack of well-powered, publicly available family-based GWAS summary statistics, we applied our prototype procedure to published results on height constructed from standard GWAS population effects[14]. When using an RDR estimate of heritability[44], $h^2_f = 0.55$, results fit predictions from the model with AM without IGEs: the predicted ratio between direct and population effects was 0.89 (S.E.=0.03), close to the observed $\delta_{\text{PGI}}/\beta_{\text{PGI}}$ =0.91 (S.E.=0.01); and the estimated contribution from IGEs was not significant. We adjusted h^2_f for AM, giving h^2_{eq}=0.70 (S.E.=0.08).

C.3.5. Planned extensions to the theoretical framework. First, we will derive theory that gives the correlations between different relative classes (e.g. *k*th degree cousins and ancestor-descendant relations) due to DGE and IGE components. This will enable us to examine the degree to which the DGE and IGE components that we estimate using the above procedure are able to explain observed relative correlations. Second, we will integrate insights from applying the xAM simulations developed in Aim 2. This will guide the development and testing of theory and methods that enable us to analyze cross-trait scenarios, such as using a PGI for one trait to predict another trait. We will apply this to, for example, EA PGIs, where there are differences between direct and population effects of the EA PGI on a wide range of biomedical and behavioral outcomes[14]. Third, we will extend our method to incorporate measured parental phenotypes as mediators of parental IGEs. Many IGE studies included measures of parental education and socioeconomic status as a form of mediation analysis[16,21]. However, these analyses did not account for confounding from AM, as our extension will.

C.3.6. Limitations of two-generation approaches. The two-generation approach works best for powerful PGIs derived from unbiased DGE estimates — as could be obtained from large-scale family-based GWASs — that are not currently available for most traits. However, PI Young and co-I Benjamin have been leading an international consortium that is producing powerful family-based GWAS summary statistics (funded by R01 AG083379-01; PI: Young, Co-I: Benjamin). The first iteration of this project is near completion and will provide sufficiently well-powered DGE PGIs for the method outlined above. We also expect that more genetic data on families and family-based GWAS summary statistics will become available during this project, including for multi-ancestry samples. Another limitation is the models we have developed do not include population stratification[4]. However, the three-generation genetic analysis methods we develop below address this issue.

C.3.7. Development of three-generation genetic analysis methods. Here we develop methods to enable three-generation genetic analysis methods that overcome some of the limitations of the two-generation methods developed above. The reason three-generation models are useful is because we can use the randomization of genetic material from grandparents to parents to remove confounding from population stratification and AM in estimates of parental IGEs. A simple extension of the two-generation model adds all four grandparents' PGIs to the regression. The expectation of a parent's PGI given the parent's parents' (grandparents') genotypes is simply the average of the grandparents' PGIs, with variation around this expectation due to random segregations during meiosis in the grandparents. This implies that by fitting the three-generation model, the information used to estimate the coefficients on the parents' PGIs is independent of population stratification — just as controlling for parental PGIs in the two-generation model removes population stratification from estimates of the direct effect. Controlling for grandparental PGIs also removes most of the confounding due to AM since much of this confounding comes from correlations between alleles in the grandparents that are not present in the random variation in the parents' PGIs due to meiosis in the grandparents. However, some confounding due to AM can remain when the parental PGIs have causal effects on mate choice, as expected when the PGI captures DGEs on the phenotypes on which parents match. We will demonstrate this in simulations below.

The major impediment to fitting three-generation models is the lack of fully genotyped three-generation pedigrees. However, we can use the imputation methodology developed by PI Young[4] — implemented in *snipar* — to impute the missing grandparental genotypes. The imputation proceeds by applying Mendelian laws to figure out which alleles of un-genotyped parents have been observed in a nuclear family. In the context of imputing missing grandparental genotypes, one can use, for example, a father and a father's sibling to impute the missing paternal grandparents' genotypes. Alternatively, with a father and one paternal grandparent genotyped, *snipar* can impute the missing grandparent's genotype. The imputed grandparents' PGIs can then be used in place of the true grandparental PGIs without introducing bias into the coefficient estimates, provided that there is not strong population structure in the sample[4,102]. Even if a parent does not have genotyped relatives, we extend the existing imputation methodology to impute the missing grandparents' PGIs (linearly with an AM adjustment), increasing power even though these samples do not contain any identifying information on their own[102].

We also investigated using uniparental three-generation regressions as this enables all samples with a genotyped father/mother to be used to estimate paternal/maternal IGEs, as opposed to only samples with both parents genotyped, which would be required to fit the biparental three-generation regression. For example, the paternal three-generation regression is: $Y_{ij} = \gamma \text{PGI}_{ij} + \eta_{\text{PGI}:p}\text{PGI}_{p(i)} + \alpha_{\text{PGI}:pp}\text{PGI}_{pp(i)} + \alpha_{\text{PGI}:mp}\text{PGI}_{mp(i)} + \epsilon_{ij}$; where $\text{PGI}_{pp(i)}$ and $\text{PGI}_{mp(i)}$ are the paternal grandfather and grandmother's PGIs. We have changed the coefficient on the offspring PGI to γ because it cannot be interpreted as a direct effect without fitting both parents' PGIs in the regression. Since the goal of the three-generation model is to enable robust inference of parental IGEs, the fact that we do not obtain an unbiased estimate of the direct effect from this regression is irrelevant.

C.3.8. Preliminary three-generation results. We simulated genotype-phenotype data for 30,000 families for traits affected by DGEs and AM, with parents' phenotypes correlated at 0.5. This corresponds to the type of AM that can generate a bias in three-generation models. We simulated 20 generations of AM except for one trait where we simulated one generation of AM for a non-equilibrium scenario. We examined PGIs with weights given by DGEs plus different levels of noise, simulating GWAS sampling error in effect estimates (Table 1).

As expected (Figure 3), the two-generation model gives statistically significant estimates of the average coefficient on the parental PGIs (average NTC) except when the PGI explains all the heritability, an unrealistic scenario that we subsequently ignore. The biparental three-generation model gave smaller average coefficients on parental PGIs, except in the non-equilibrium case. However, uniparental three-generation regressions gave smaller coefficients on the parents' PGIs than the biparental three-generation regression. This indicates that uniparental three-generation regression has smaller bias due to AM as well as increased power — due to less stringent data requirements — over the biparental three-generation regression. We found similar results using grandparental PGIs imputed from parents and their siblings' genotypes in place of true grandparental PGIs.

C.3.9. Extensions of three-generation approaches. We will extend the theoretical framework developed above to analyze uniparental three-generation regressions. This will enable adjustment for the residual bias due to AM present in three-generation regressions. Furthermore, three-generation data presents an opportunity to investigate mechanisms of AM. By using random, within-family variation in the parents' PGIs, we can measure the degree to which PGIs capture DGEs on mate-choice. We will decompose the correlation between parents' PGIs into a component due to DGEs on mate choice and a residual component explained by other factors such as population structure. This will help advance our limited understanding of mechanisms of AM.

	Fraction of h2 explained (k)		Avg. NTC (two-gen)		biparental		paternal		maternal		biparental imputed		paternal imputed		maternal imputed	
PGI	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**	**Est.**	**S.E.**
DGE	0.996	0.013	0.000	0.003	0.000	0.004	-0.005	0.005	0.006	0.005	0.000	0.004	-0.006	0.006	0.006	0.006
DGE + noise (1)	0.480	0.010	0.076	0.003	0.028	0.005	0.011	0.006	0.014	0.006	0.031	0.005	0.015	0.007	0.016	0.007
DGE + noise (10)	0.070	0.004	0.071	0.004	0.026	0.005	0.015	0.007	0.004	0.007	0.018	0.006	0.009	0.008	-0.005	0.008
DGE + noise (1) + non-eq.	0.577	0.011	0.029	0.004	0.029	0.004	0.028	0.006	0.018	0.006	0.025	0.005	0.025	0.007	0.014	0.007

Table 1. Preliminary results from three-generation models. *Each row gives a PGI with a different level of noise in the weights, with all scenarios at approximate equilibrium apart from the last row. We give the fraction of heritability the PGI would explain in a random-mating population, k, as estimated by our procedure (Figure 4); followed by the average non-transmitted coefficient (NTC) from fitting a two-generation model; then we give the average coefficient on the parents' PGIs from a biparental three-generation model; then the coefficients on paternal and maternal PGIs in uniparental three-generation regression; followed results from the same three-generation models but using grandparental PGIs imputed from parents and their siblings' genotypes.*

C.3.10. Proposed multi-ancestry applications. We propose to apply our two-generation methods to <u>public datasets</u> with sufficient numbers of genotyped sibling and/or parent-offspring pairs to enable powerful two-generation PGI analysis. For European ancestry samples, we will apply our methods in UK Biobank[103], Finngen[104], Generation Scotland[105], and the Millennium Cohort Study[106]. PI Young has access to individual level data and approval for performing polygenic prediction analyses in these datasets. We will analyze the Mexico City Prospective Study (MCPS)[31], whose ancestry is mixed primarily between European, Indigenous American, and African genetic ancestries. PI Young and consultant Sohail have made an application to perform family-based genetic association analyses in MCPS. Access to MCPS is granted on a similar basis to UK Biobank, so we anticipate that our proposal will be approved. Analyses will be done by consultant Sohail and her group during the period in which access is only granted to researchers in Mexico. After this expires (~2 years), PI Young and his team at UCLA will be able to access the individual-level data to perform further analyses. We will analyze the Taiwan Biobank — through the access co-I Zaitlen has — to perform the first robust and powerful estimation of IGEs in an East Asian genetic ancestry sample.

Dataset	Par. / off. pairs	Sib. pairs	Father w/ sib or par.	Mother w/ sib or par.
Finngen (R12)	100k	92k	16k	20k
UK Biobank	6k	22k	0k	0k
Generation Scotland	10k	8k	2k	4k
Millennium Cohort Study	12k	0k	0k	0k
Mexico City Prospective Study	32k	29k	?	?
Taiwan Biobank	7k	7k	?	?

Table 2*. Genotyped sample sizes for family-based analyses rounded to closest 1000.*

<u>The datasets we will analyze provide a wide range of phenotypes, including anthropometric, metabolic, psychiatric, educational, cognitive, behavioral, and electronic health record derived disease phenotypes.</u> We will apply our two-generation analyses to family-based GWAS derived PGIs from the meta-analysis project led by PI Young, which will produce summary statistics for 34 phenotypes including EA, cognitive ability, height, BMI, age at first birth, depression, and blood pressure and lipids.

We anticipate that there will be more than sufficient power to perform two-generation analyses in all datasets since only a few thousand sibling and/or parent-offspring pairs are required to obtain precise estimates of the two-generation model[16,21]. For three-generation analyses, we anticipate having sufficient power for those datasets with large numbers of genotyped parent-offspring pairs, where those parents also have a genotyped sibling or parent (Finngen and Generation Scotland, Table 2). It may be possible to apply three-generation methods in MCPS due to the household level sampling of this study[106], but the relevant sample sizes have not been published. For three-generation analyses, there is no assumption that the PGI is derived from family-based GWAS, so we will apply our methods to powerful PGIs derived from the PGI repository[107], on which co-I Benjamin is the PI. We will examine evidence for IGEs of PGIs for ADHD, age at first birth in women, BMI, depression, EA, ever-smoker, height, and number of children.

C.4. Aim 2. Advanced simulation tools and open simulation library and benchmarking infrastructure.

We propose developing two complementary simulation tools (Table 3) capable of a broad array of phenotypic and genetic architectures incorporating intergenerational dynamics, including both xAM and vertical transmission (VT). When a heritable parental phenotype affects the offspring through the environment (VT), this induces IGEs. These intergenerational dynamics have largely been overlooked in the development of human genetics methods. Further, we propose to create an open simulation library and benchmarking platform to facilitate the development of methods that can identify meaningful biological signals in realistically complex, structured population datasets.

C.4.1. Advanced multivariate forward time simulation of complex intergenerational dynamics. We propose to develop *xftsim*, a powerful forward time simulation Python package enabling efficient simulation of high-dimensional phenotypes with complex mating, intergenerational transmission, and spatial dynamics using biobank-scale marker data. <u>Prototype implementations of several major features demonstrate initial success</u> (Table 4). <u>*Memory-efficient representation of genetic data*</u>. Given the high-dimensionality of genotype data, memory management is essential to feasibility. We propose to represent haplotypes as labeled arrays of 8-bit integers with lazy evaluation backed by distributed files. Phased haplotype VCFs are converted to *zarr* file format[108], which allows specific rows or columns to be accessed in parallel with minimal overhead. These files are "lazily" represented as *Dask* arrays[109]; elements can be manipulated without being loaded into memory until computations are invoked. In the course of a forward time simulation, only haplotypes at causal loci ever need to be accessed (for, e.g., computing phenotypes), except when writing out simulation results. This means that only a fraction of the genotype data needs to be loaded into memory. We have previously used this approach to simulate univariate AM with 1 million causal variants in a founder population of 3 million individuals[73].

Feature	*Forward-time simulation (Aim 2.1)*	*"One-shot" simulation (Aim 2.2)*
Computational efficiency	Low-memory footprint	Linear in sample / genome size
Dependence on dynamical theory	Mathematical dynamics need not be known	Theoretical results required
Complex additive genetic architecture	Yes	Yes
Non-additive genetic architecture	Yes	No
Complex mating patterns	Yes	Yes
Complex local LD / recombination	Yes	No
Vertical transmission	Yes	No
Disequilibrium	Yes	No
Auxiliary -omics	Yes	No
Reproducible / Open	Yes	Yes
Summary	*Maximally expressive, theory agnostic*	*Maximally efficient, exploits theory*

Table 3. *Features of forward-time and one-shot simulation approaches.*

<u>*Memory-efficient representation of meiosis and genetic transmission*</u>. Meiosis and genetic transmission are central to

forward-time simulation and we use three approaches for different scenarios, two of which we have explored in previous work. First, when the data are low-dimensional (e.g., ≤128k individuals by ≤16k variants), we store the entire haplotype matrix in memory[78]. In this case, we simulate recombinations as a random process on a genetic map, shuffling subsets of rows of the haplotype matrix. Second, when high-coverage output is desired (i.e., sequence data) but realistic recombination is not a concern, we chunk the entire genome into a fixed number of blocks and restrict recombination events to the boundaries of these blocks[73]. At the end of simulation, full sequence data can be reconstructed from the block representation in parallel across multiple devices, maintaining a low memory footprint. Finally, when both high-coverage and realistic recombination are paramount, we propose to draw recombination events continuously along the genome and track contigs using tree sequences, together with a map between these sequences and founder haplotypes[110]. This approach, similar to that used by Tahmasbi and colleagues in *GeneEvolve*[36], enables realistic recombination while minimizing short-term memory footprint. However, by using tree sequences rather than intervals corresponding to contiguous regions of the genome, our approach avoids exponentially growing memory requirements. Recombination will support ancestry-specific maps (e.g., from the pyrho project[111,112]).

	Feature	*Status*	*Approach*
a1	Multivariate assortative mating	Part.	Linear approach for univariate AM / rank-one xAM, quadratic mixed integer program approach for arbitrary high-dimensional xAM with batching for efficiency implemented; alternative approaches (e.g., agent-based modeling) to be incorporated via existing arbitrary simulation state → mate pairing map abstraction
a2	Vertical transmission / direct indirect effects	Impl.	Arbitrary causal relationship between phenotype components across generations
a3	Complex genetic / architectures	Part.	Parametric infinitesimal and non-infinitesimal multivariate additive architectures implemented, arbitrarily complex architectures possible by manually specifying effects but will extend to incorporate parametric LD/MAF dependent architectures and annotation-specific effects (*d3*)
a4	Nonlinear genetic effects	Part.	Quantitative gene-by-context (GxC) (i.e., true polygenic index by context) interaction implemented, will extend to locus-level GxC and GxG interactions (including special case of dominance), coordinated epistasis
a5	IGEs among siblings	Prop.	Intragenerational dependencies between sibling phenotypes
a6	Variable population size	Prop.	Arbitrary distributions of number of mates / offspring per individual, allowing for dependence on phenotypes (i.e. natural selection)
a7	Sub-populations / admixture	Prop.	Discrete founder sub-populations with phenotypic (*a1*) or spatial (*d4*) mating constraints; population-specific recombination maps
Data modalities			
d1	In-memory haplotype array	Impl.	Array of int8 with lazy evaluation via *Dask* and variant-level (e.g., ref/alt alleles, AF, rsid, position, etc.) and individual-level (e.g., individual / family ID, sex, etc.) stored as *xarray* coordinates
d2	Larger-than-memory haplotype arrays	Prop.	*Dask* / *Xarray* as above backed by *zarr* files to enable loading sparse variant sets (e.g., causal variants) with minimal memory footprints, together with tree sequence-to-array coordinate map
d3	Biological annotations	Prop.	Incorporated via *xarray* coordinates along variant dimension
d4	Geographic / spatial phenotypes	Prop.	Phenotypes represented as two-dimensional *xarray*s that can be projected onto real world geography, migration modeled either by discretized vector field of flows or via mating scheme (a1)
d5	Auxiliary omics / imaging data	Prop.	Supported in principle via existing arbitrary simulation state → phenotype map abstraction

Table 4. Architectural / dynamical phenomena modeled by *xftsim*. *Note*: *“Impl.” = prototype implemented, “Part.” = prototype partially implemented, “Prop.” = proposed*

<u>Modeling complex architectures and intergenerational transmission</u>. We propose to define architectures at the level of *phenotypic components* (e.g., the contributions of additive genetic liability, spatial environment, or rearing environment to a trait) rather than aggregates phenotypes, making it simple to incorporate intergenerational transmission, interactions and causal dependencies between components of multiple phenotypes, and numerous other complex architectures. Genetic components can be arbitrary user-defined functions of haplotypes, enabling simulation of dominance, epistasis, and gene-by-context interactions, though we will provide convenience functions for generating causal effects according to commonly used models[22,24,113,114].

<u>Modeling complex mating patterns</u>. Mating regimes will be defined as arbitrary maps from simulation state (including, e.g., offspring and parental phenotypic components) to mate pairings and offspring counts, enabling xAM across spatial, environmental, genetic, and/or individual-specific phenotype components. While this general framework enables users to implement arbitrary mating regimes of interest (e.g., using agent-based approaches), we implement two broadly applicable regimes (in addition to random mating). *Linear xAM* efficiently sorts and pairs individuals using a linear combination of phenotypic components and randomly generated noise. However, we have observed that several dimensions are required to explain cross-mate similarity across a broad array of phenotypes in the UK Biobank (Figure 1b). As a result, we have formalized and solved the *general xAM* problem of mating n pairs of individuals across k phenotypic components, represented respectively by the n-by-k matrices $\boldsymbol{Y}$ (females' phenotypes) *and* $\boldsymbol{Z}$ (males' phenotypes), to achieve a user-specified cross-mate (potentially asymmetric across sexes) cross-correlation matrix $\boldsymbol{\Omega}$ as the following quadratic integer program (QIP): $\text{minimize}_{\boldsymbol{P}\in\mathcal{P}_n}\|\boldsymbol{Y}^T\boldsymbol{P}\boldsymbol{Z}-\boldsymbol{\Omega}\|_F$ where $\mathcal{P}_n$ is the space of n-dimensional permutation matrices (corresponding in this case to the set of potential mate pairings). This QIP is equivalent to a long-standing NP-hard combinatorial optimization that in practice can be approximated efficiently[115]: in our prototype implementation, 500 mate pairs across five dimensions takes approximately five seconds. Processing mate pairs in batches of 500 is sufficient for achieving arbitrary cross-mate cross-trait cross-correlations.

<u>Preliminary results</u>. Using a prototype of *xftsim* with partial functionality (Table 4) we simulated genetically independent phenotypes each with random mating h^2 = 0.5 over five generations of random mating (RM) or xAM across two- (2xAM) or five- (5xAM) dimensional xAM with cross-mate correlations all fixed at 0.2 across all pairs

of phenotypes, with and without multivariate VT, such that 5% of the variance of each phenotype was due to the combined effects of parental phenotypes, thus incorporating environmental transmission and IGEs. Comparing the true PGI correlations (r_{score}), and estimated genetic correlations ($\hat{r}_{\mathrm{HE}}$)[116], which are commonly interpreted as measures of pleiotropy[117–120], we make several key observations (Figure 6). First, while 5xAM leads to moderate increases in score correlation vis-a-vis 2xAM, it induces substantially larger genetic correlation estimates (50% larger after five generations), demonstrating the impact of high dimensional xAM. Second, under random mating (RM), VT induces no change in true score correlations but increases genetic correlation estimates via IGEs. Finally, though high-dimensional xAM and VT both independently induce substantial bias in genetic correlation estimates, when both dynamics are present their effects are compounded, yielding dramatically misleading estimates. These results illustrate the importance of a multifactorial approach to understanding the consequences of complex intergenerational dynamics and demonstrate the utility of the *xftsim* project for flexibly modeling such phenomena.

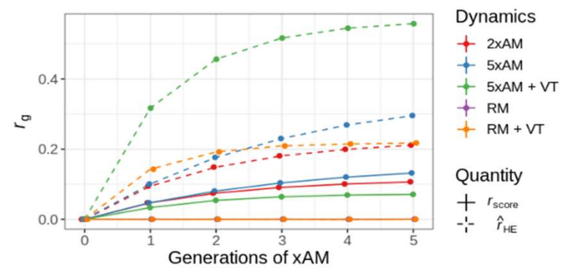

Figure 6. *Estimated genetic correlations (dashed lines) and true PGI correlations (solid) across simulations of complex intergenerational dynamics.*

C.4.2. Efficient "one-shot" genotype / phenotype simulation and theory in multivariate contexts. Forward time simulation comes at the cost of simulating each generation. One-shot approaches leverage theoretical results to avoid unused intermediate generations, achieving substantial computational gains (Figure 7a). Co-I Border and colleagues developed the *rBahadur* R library[40] for efficient simulation of univariate AM at equilibrium, i.e., the steady state achieved after ≈10 generations of assortment (Figure 7b). This approach is based on three insights. First, classical results regarding the equilibrium distribution of unlinked causal variants under AM[42] can be reformulated to approximate the haploid LD correlation matrix as a rank-1 perturbation of a diagonal matrix: $\boldsymbol{R}_{AM} := \boldsymbol{D} + \phi\phi^{\mathsf{T}}$ where $\boldsymbol{D}$ and ϕ are both described by known algebraic functions that operate elementwise on the vector of standardized causal effects and are parameterized by the random-mating heritability and cross-mate phenotypic correlation[73]. Second, the AM-induced correlations among causal variants are small enough that the joint distribution of haploid SNPs is well approximated by the Bahadur order-2 multivariate Bernoulli (MVB) distribution[121,122]. Third, the diagonal-plus-low-rank (DPLR) approximation of R_{AM} can be used to sample from this distribution with linear time complexity (Figure 7a), versus the quadratic or worse complexity required to sample from arbitrary correlation structures[40]. We propose to generalize classical theoretical results to multivariate xAM and disequilibrium, both of which preserve the DPLR structure, enabling incorporation into the *rBahadur* library to give rapid one-shot simulation of genotype & phenotype.

<u>*Extension to multivariate xAM*</u>. In preliminary work, we have proven that for a moderately polygenic trait, in analogy with the univariate case, the equilibrium correlation structure for m unlinked haploid causal variants affecting one or more of k phenotypes is well approximated by $\boldsymbol{R}_{xAM} := \boldsymbol{D} + \boldsymbol{\Phi}\boldsymbol{\Phi}^{\mathsf{T}}$ where $\boldsymbol{\Phi}$ is now an m-by-k matrix-valued function of the m-by-k matrix of causal effects $\boldsymbol{\beta}$ parameterized by the random-mating genetic covariance matrix and cross-mate cross-trait phenotypic correlation matrix. The functional form of $\boldsymbol{\Phi}$ has not been previously characterized. We have proven that, for polygenic traits, $\boldsymbol{\Phi}$ is well approximated by $\boldsymbol{\beta\Lambda}$, where $\boldsymbol{\Lambda}$ is a k-by-k matrix. We propose to apply the approach of Nagylaki[42] to determine the precise functional form of $\boldsymbol{\Phi}$, but failing that, we can employ the linear approximation of $\boldsymbol{\Phi}$. Preliminary simulation work has demonstrated that employing a linear approximation yields equivalent results in terms of true equilibrium heritability and multiple heritable estimators. We will enable users to generate genotype-phenotype data given the cross-mate correlation matrix, allele frequencies, and either effect sizes or the random mating genetic covariance matrix as input.

<u>*Extension to disequilibrium (x)AM*</u>. Nagylaki[42] characterized the joint equilibrium distribution of causal haplotypes under univariate AM in two stages using elementary methods. First, he derived recurrence relations for the pairwise cis- and trans-haploid correlations (i.e., population correlations among alleles inherited from the same parent, or different parents, respectively) and cross-mate haplotypic correlations based on Mendelian inheritance. Then, by solving the system that results at equilibrium when the cis-, trans-, cross-mate correlations are equal and relating these solutions to summary statistics with well characterized dynamics

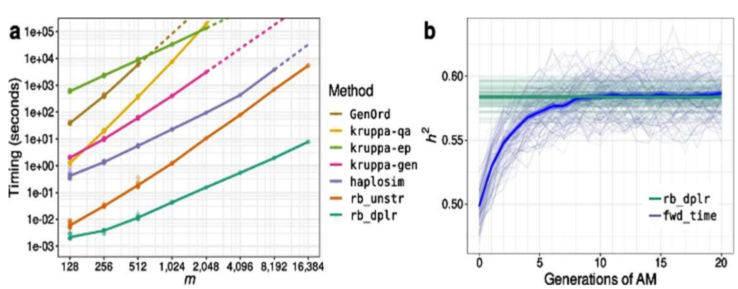

Figure 7. *(a) The rb_dplr algorithm has linear time complexity in sample size and number of causal variants; all other methods have quadratic time complexity or worse. (b) Preliminary generation of genotype & phenotype data recapitulates forward time simulation with less variance*[40]*.*

(i.e., the genetic variance and cross-mate genetic covariance) he derives approximations of the pairwise haploid correlations. We propose to use the same recurrence relations, together with an augmented set of summary statistics that vary across generations, to characterize the haploid correlation structure after an arbitrary number of generations. In preliminary work, we have fully characterized the dynamics of an alternative set of summary statistics, compromised by the effect-size weighted pairwise sums of trans-, cis-, and cross-mate haploid correlations. We propose to approximate correlations at the level of pairs of loci using these results together with monotonicity constraints (e.g., we expect haploid correlations involving loci with larger effects to be larger in magnitude). In addition to providing further insights into the dynamics of causal genotypes subject to AMs, this will enable us to approximate the generation *t* diploid correlation matrix, again as a low-rank perturbation of a diagonal matrix, capturing differences in trans- versus cis-LD. We will implement this to allow for simulation of genotype-phenotype data under disequilibrium AM after a specified number of generations of AM with potentially time-varying cross-mate correlations. Further, we will extend this approach to multivariate xAM by applying the techniques described above to characterize the joint distribution of haplotypes under high-dimensional disequilibrium xAM and enable efficient simulation via the *rb_dplr* algorithm.

C.4.3. Open simulation / benchmarking library for investigating impacts of complex intergenerational dynamics on polygenic estimators. Research progress in a number of fields, including computer vision[130]and population genetics[131,132], has benefited from the establishment of open training datasets and simulation libraries. We propose to build an open simulation library and benchmarking platform for the statistical genetics community (*sgbench*), enabling researchers to compare existing and novel methods across a comprehensive array of realistic architectures and intergenerational dynamics and to identify weaknesses in current approaches. We propose to develop *sgbench* in three phases, initially focusing on realistic dynamics & genetic architectures that complicate interpretation of genetic association statistics and potentially introduce bias into variance components estimates and PGIs (Table 6). In Phase I, we will explore a broad array of architectures using simplistic, synthetic genetic data and identify a smaller set of architectures sufficient for capturing variability in estimator performance. In Phase II, we will rerun this limited set of simulations using realistic genotype data, which together with the benchmarking platform, will form the initial *sgbench* release. In Phase III, we will periodically update the simulation library as necessary and develop additional simulation modules focused on specialized analyses.

Polygenic estimators	*Focal statistics*
GWAS w/ linear regression[123] / linear mixed models[124] / sib-difference[8]	bias, power, false positive rate (FPR), colocalization power, colocalization FPR
REML / MAF-LD partitioned REML[27,39]	Heritability / genetic correlation estimates / enrichments
LDSC / MAF-LD partitioned LDSC[22,125]	
Relatedness disequilibrium regression[44]	
Cross-validated PGI[126]	Estimated PGI correlation, cross-mate PGI correlation
Methods developed in Aim 1	Contribution of IGEs, cross-mate genetic correlation, equilibrium heritability

Table 5. *sgbench initial release estimators.*

<u>Exploratory Analysis (Phase I)</u>. In Table 6 we enumerate an extensive set of phenomena previously demonstrated to be relevant to human complex traits: considering all combinations yields 216 distinct scenarios. We propose to reduce this number to the greatest extent possible by identifying the minimal set of realistic scenarios necessary to capture key patterns of bias within and across family- and population-based methods. To do this we will run simplified versions of all 216 simulations enumerated in Table 6 using 400,000 founders with synthetic genotypes at 32,000 loci (including 2,000 causal loci) generated using the *HAPNEST* software[37], performing 100 replicates per scenario, applying estimators within the simulation framework to minimize storage requirements. We estimate that this will take approximately two days using 1% of the cores on the UCLA Hoffman2 compute cluster. We will then apply sparse nonlinear canonical correlation analysis to define a minimal subset of no more than 50 scenarios necessary to capture variation in estimator performance[133,134]. We will also determine the number of simulation replicates required to reliably assess estimator performance.

<u>Initial release (Phase II)</u>. To form the initial release, we will rerun the minimal subset of simulations using denser genotype data. All simulations will draw from a pool of 1.6 million synthetic founder genotypes at approximately 1.1 million common HapMap3 loci[135], divided equally across four genetic ancestry groups[136] (AFR, AMR, EAS, EUR) generated using *HAPNEST*[37]. Because our forward time simulation approach only requires loading causal variants into memory and tracks recombination events using compact tree sequences, the increased genotype density will not increase computational burden. From each simulation

Feature	*Variants*	N_{sim}
Effect architecture	MAF-LD independent / dependent[24,27,114]	2
Pleiotropy	Correlated / non-overlapping[78]	2
Missing heritability	None / 50% CVs (rare) omitted[127]	1*
Mating	Random / linear 5xAM[2] / spatially constrained[9]	3
GxC	None / true PGI by environment[128]	2
Epistasis	None / Coordinated / Uncoordinated[129]	3
Population structure + environmental stratification	None / spatially varying[78]	2
Vertical transmission	None / phenotypic / environmental[21]	2

*Missing heritability can be simulated by simply omitting genotypes
Note: The total number of scenarios is the product of N_{sim} values

Table 6. *Phase I dynamics and architectures.*

run, we will produce three datasets for benchmarking purposes: a set of 48,000 "unrelated" samples (selectively pruning away third-degree or closer relatives); a set of 48,000 nuclear families, each including two siblings and their parents; and a set of 48,000 extended pedigrees spanning three generations. We will produce genotypes at 64,000 common loci (minor allele frequency [MAF] > 0.01) distributed across the genome, including 4,000 causal variants. These numbers were chosen based on preliminary investigations to balance three requirements: 1) limiting computational burden of benchmarking, 2) generating marker data with sufficient complexity (i.e., variable local LD and MAF), and 3) sufficiently large sample sizes to evaluate family-based approaches. The entire simulation library and all data will be available through publicly accessible cloud-based storage. To maximize ease-of-use and minimize storage burden, simulations will be accessible in multiple formats, through multiple protocols: users may 1) use cloud compute without downloading any data locally; 2) download individual datasets and run benchmarks locally (we estimate that the unrelated individual data will approximately 2GB per dataset); 3) download founder data and tree sequences to reconstruct data locally; 4) recreate all data from scratch using provided scripts and random seeds, which we estimate could be done in under two days using 512 cores; and 5) download standard or family-based GWAS summary statistics and LD scores, the total size of which is small (<1TB), as these are all that is required to run many widely used methods[125,137]. Finally, we will publish our initial set of performance benchmarks on a public git repository, which we will open to user submissions.

Maintenance and extension (Phase III). We will annually update the repository with relevant novel methods published in top-tier genetics journals. Based on community demand, we will explore creating additional modules narrowly focused on particularly challenging architectures or methods requiring auxiliary data types.

C.4.4. Potential pitfalls (Aim 2)

Scalability and shareability. Depending on the scenario, computation in *xftsim* is dominated either by mate assignment or meiosis. With respect to the first, using a prototype simulation with serial batched general xAM with a population of 128,000 individuals takes approximately 22 minutes; by parallelizing across batches, we will be able to reduce this by an order of magnitude. With respect to the second, we have previously used similar data structures (i.e., Dask arrays backed by zarr files[108,109]) to simulate univariate AM in 3 million individuals tracking 1 million variants[73]. With respect to *sgbench*, the most salient scalability issue concerns the shareability of a large library of models. We address this by providing several protocols for using the benchmarking suite without downloading large amounts of genotype data.

Dependence of *sgbench* on simulation tools under development. We have working prototype simulators spanning a substantial fraction of the architectural features comprising the *sgbench* library (i.e., xAM, vertical transmission, pleiotropy, and effect distribution) using in-memory haplotype arrays. We have also previously demonstrated the scalability of lazy, distributed arrays backed by zarr files[73].

C.5. Minimizing Misapplications and Communicating Potential Harms and Benefits. Our proposed research approaches can be—and will be—applied to socially sensitive traits, such as EA. Research on the genetics of such traits is prone to misapplication, misinterpretation, and misunderstanding. To mitigate potential harms, the Social Science Genetic Association Consortium—PI Young and Co-I Benjamin are on the Steering Committee—has a policy of releasing Frequently Asked Questions (FAQ) documents for its flagship papers. The goal is to describe appropriate interpretation of the research, explain which applications are—and are not—justifiable, and discuss its social and ethical implications. The FAQs are included as supplementary materials in the papers, are posted for download alongside the paper, and are sent to journalists who inquire about the paper. The FAQs have been praised by the editors of *Nature*[138] and by the Hastings Center[139] (a bioethics research institute) as a best practice. Indeed, writing FAQs is becoming routine in the field[63,140]. We will write FAQs for all major papers resulting from this research, with input and feedback from co-Is Garrison and Panofsky.